ABNORMALITIES OF HEMOGLOBIN SYNTHESIS

Learning Objectives

At the end of the lecture student should be able to demonstrate;

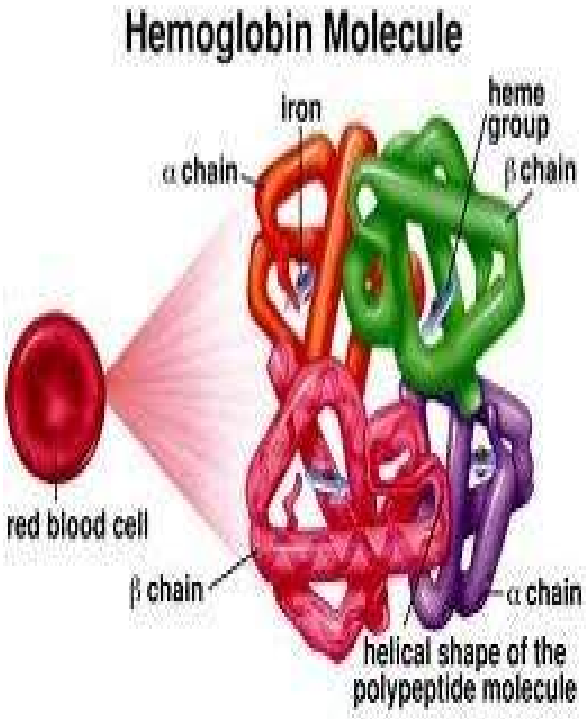
- Hemoglobin structure
- The R and T state of the hemoglobin
- The Heme structure and its synthesis
- Defects of heme synthesis
- The major forms of porphyrias
- Variants of Hemoglobin according to the chains

HEMOGLOBIN

- Roughly spherical, with a diameter of nearly 5.5nm

- **Tetrameric protein** containing **four heme prosthetic groups**, one associated with the polypeptide chain

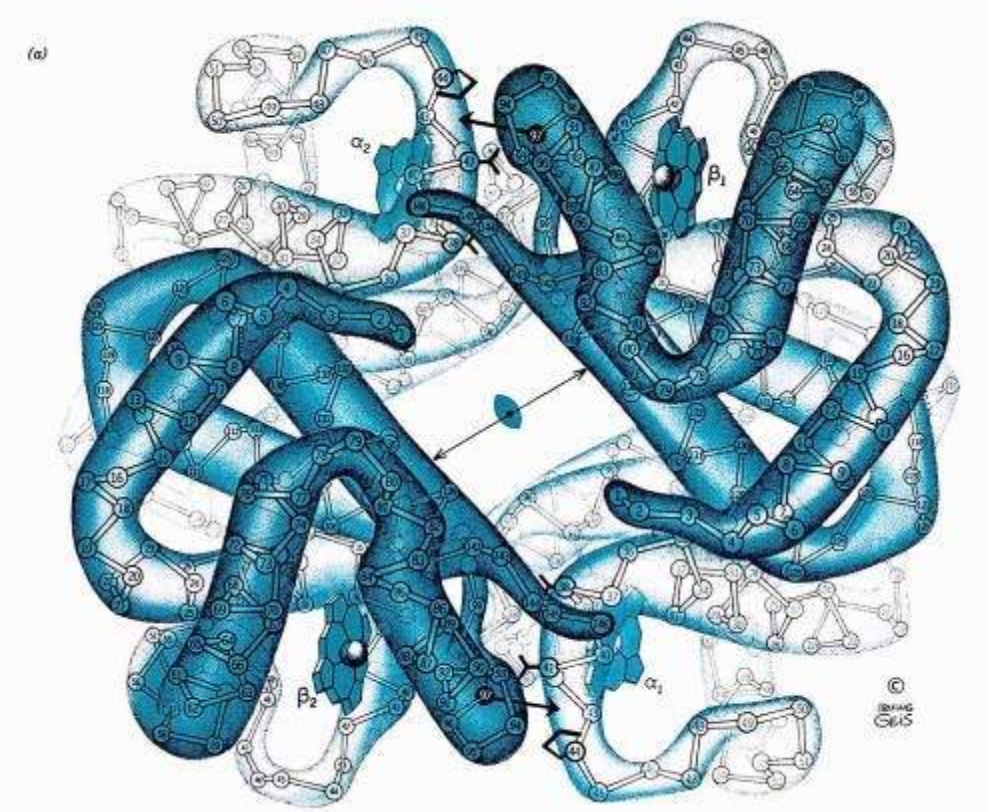
- Adult hemoglobin contains **two types of globin, two alpha (α) chains and two beta (β) chains**



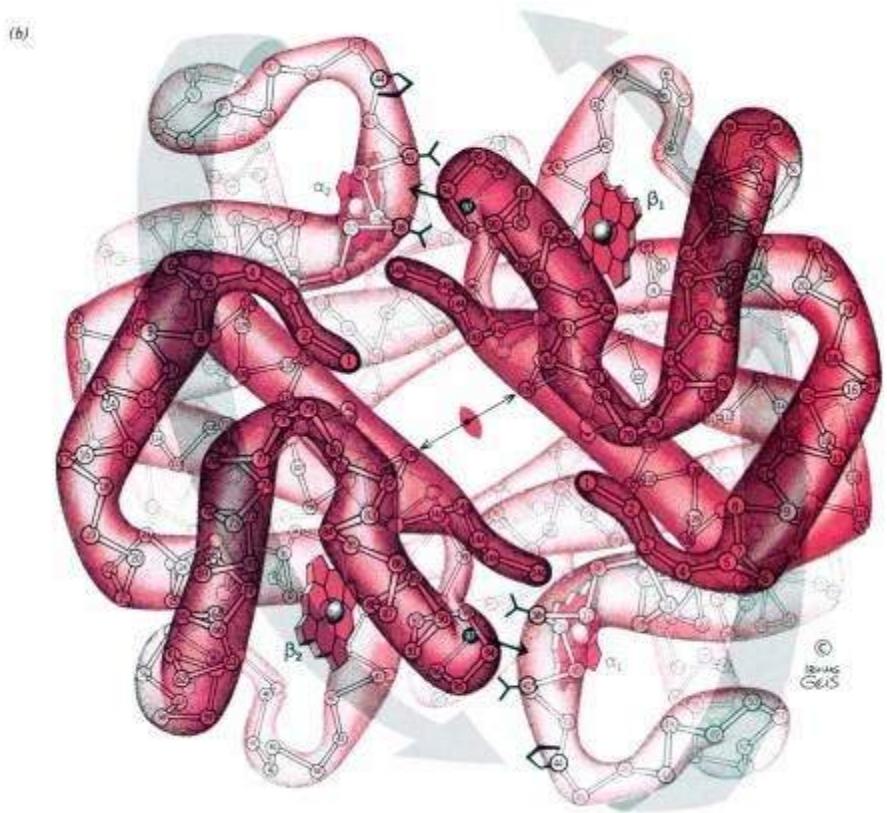
Hemoglobin undergoes structural change on binding oxygen

- X-ray analysis has revealed two major conformations of hemoglobin; the **R (relaxed) state** and the **T (tense) state**
- Although oxygen bind to hemoglobin in either state, it has a **significantly higher affinity for hemoglobin in the R state**
- Oxygen binding stabilizes the R state
- When oxygen is absent, the **T state is more stable and is thus the predominant conformation of deoxyhemoglobin**

Quaternary structure of deoxy- and oxyhemoglobin

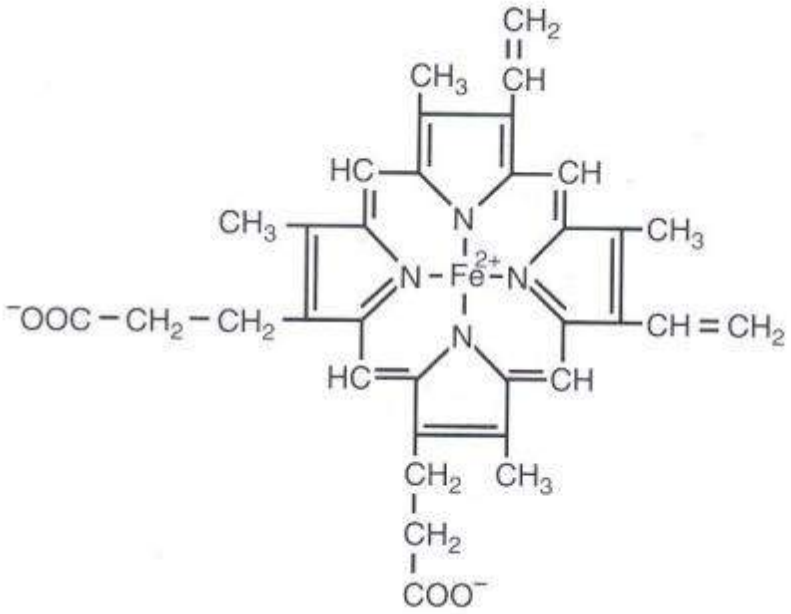


T-state



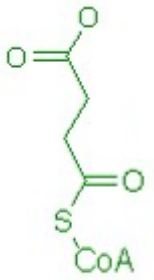
R-state

STRUCTURE OF HEME



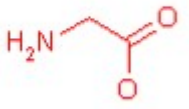
- Ferrous iron (Fe^{2+})
- Protoporphyrin IX: contains 4 pyrrole rings linked together by methenyl bridges

HEME SYNTHESIS

8 

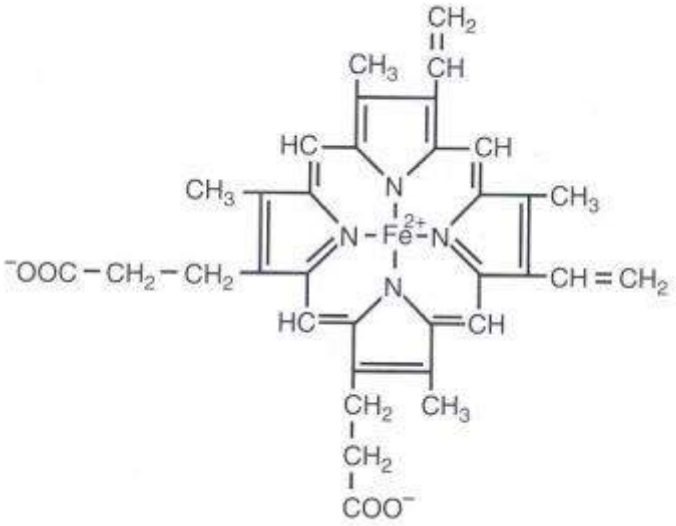
Succinyl CoA

\+

8 

Glycine





Heme

Defects of Hemoglobin Synthesis

Glycine + Succinyl-CoA

1 α-ALA synthase

δ-Aminolevulinic acid (δ-ALA)

Protoporphyrine

Fe + Transferrin

Iron deficiency etc. (see next page)

Iron deficiency anemia

Heme

Gene defect

Hereditary sidero-blastic anemia

Globin synthesis

Gene defects

Heme + 2 β-chains + 2 α-chains

1 146 1 141

Hemoglobin A_1 ($= Hb\alpha_2\beta_2$)

Hemoglobin A deficiency

α-Thalassemia (with $Hb\gamma_4$; $Hb\beta_4$)

β-Thalassemia (with Hb F $= Hb\alpha_2\gamma_2$ and Hb $A_2 = Hb\alpha_2\delta_2$)

Gene defect

2 5 Glu 7

5 Val 7

6

False β-chain

Hemoglobin S

Normal erythrocyte

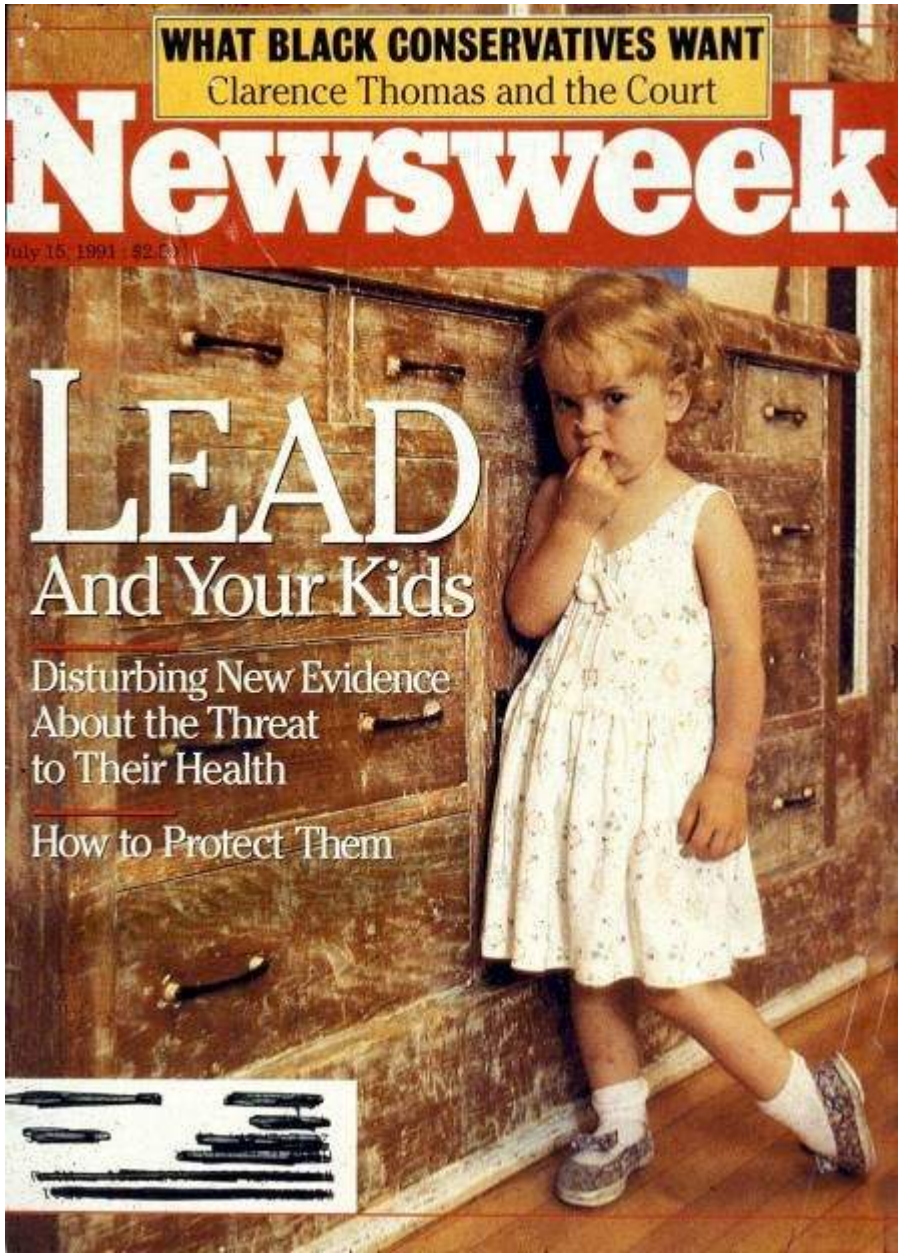
Sickle erythrocyte

Sickle-cell anemia

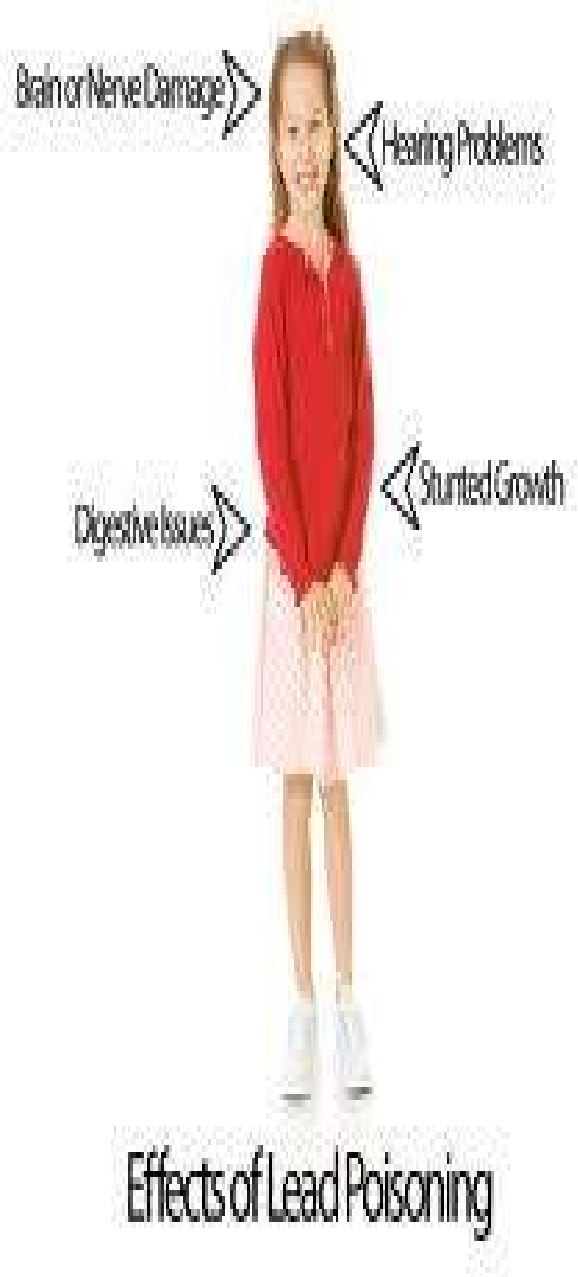
DEFECTS OF HEME SYNTHESIS….

- **Acquired: Lead poisoning**
- **Congenital: Porphyrias**

LEAD IN HOMES



LEAD TOXICITY



Symptoms

- Irritibility
- Lethargy
- Sleeplessness
- Headaches
- Poor appetite
- Abdominal pain (with or without vomiting)
- Constipation

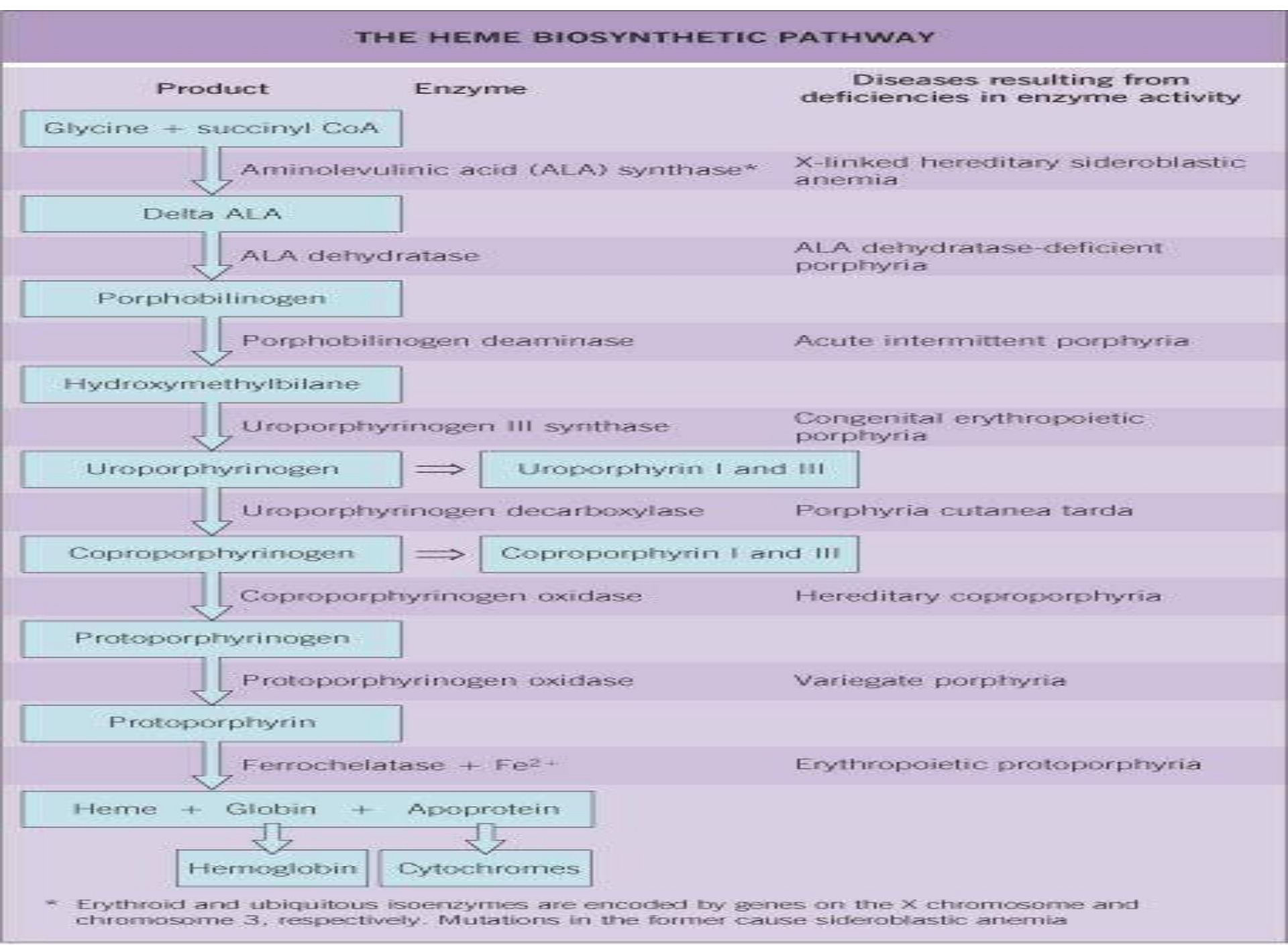
Pathophysoiology

- Binds to any compound with a sulfhydryl group
- Inhibits multiple enzyme reactions including those involved in heme biosynthesis (PBG synthase & ferrochelatase)



PORPHYRIAS

- **These are a number of inherited (genetic) diseases which result in excretion of large amounts of porphyrin precursors.**


THE HEME BIOSYNTHETIC PATHWAY
Product
Enzyme
Diseases resulting from deficiencies in enzyme activity
Glycine + succinyl CoA
Aminolevulinic acid (ALA) synthase*
X-linked hereditary sideroblastic anemia
Delta ALA
ALA dehydratase
ALA dehydratase-deficient porphyria
Porphobilinogen
Porphobilinogen deaminase
Acute intermittent porphyria
Hydroxymethylbilane
Uroporphyrinogen III synthase
Congenital erythropoietic porphyria
Uroporphyrinogen
Uroporphyrin I and III
Uroporphyrinogen decarboxylase
Porphyria cutanea tarda
Coproporphyrinogen
Coproporphyrin I and III
Coproporphyrinogen oxidase
Hereditary coproporphyria
Protoporphyrinogen
Protoporphyrinogen oxidase
Variegate porphyria
Protoporphyrin
Ferrochelatase + Fe2+
Erythropoietic protoporphyria
Heme + Globin + Apoprotein
Hemoglobin
Cytochromes
* Erythroid and ubiquitous isoenzymes are encoded by genes on the X chromosome and chromosome 3, respectively. Mutations in the former cause sideroblastic anemia

CLASSIFICATION OF THE PORPHYRIAS

Acute porphyria	Non-acute porphyria
ALA-D deficient porphyria	Porphyria cutanea tarda
Acute intermittent porphyria	Erythropoietic protoporphyria*
Variegate porphyria	Congenital erythropoietic porphyria*
Hereditary coproporphyria	

1) ***ACUTE INTERMITTENT PORPHYRIA:***

- Due to deficiency of uroporphyrinogen1 synthetase
- Patients excrete large amounts of porphobilinogen & δ-aminolevulinate in urine.
- Urine darkens on exposure to light
- Patients are not photosensitive
- Symptoms include abdominal pain, vomiting, constipation & neuropsychatric signs.

2) PORPHYRIA CUTANEA TARDA:

- Due to deficiency of uroporphyrinogen decarboxylase
- Urine contains large amounts of uroporphyrins I & III
- Patients are photosensitive
- The most common porphyria

3) HERIDITARY COPORPHYRIA:

- Due to deficiency of coporphyrinogen oxidase
- Coporphyrinogen III and other intermediates prior to block accumulate in urine
- Patients are photosensitive

4) GENETIC ERYTHROPOIETIC PORPHYRIA:

- Due to deficiency of coporphyrinogen III cosynthase
- uroporphyrinogen I & coporphyrinogen I accumulate in urine
- Patients are photosensitive

HEMOGLOBIN VARIANTS

Type	Structure	Comments
HbA (95%)	$\alpha_2\beta_2$	
HbA_2 (4%)	$\alpha_2\delta_2$	Functionally, this variant is indistinguishable from HbA Mutations in δ-globin are without effect
HbF (1% in adults – predominate form in the fetus during the 2nd and 3rd trimesters of pregnancy)	$\alpha_2\gamma_2$ His^{143} (β) → Ser (γ)	Interaction with 2,3-BPG is weaker resulting in an increased affinity for O_2 and a greater stabilization of the R state. This allows for a more efficient transfer of O_2 from maternal to fetal hemoglobin

HEMOGLOBINOPATHIES

> 600 mutations: several hundred of these result in a pathological phenotype

Examples and classification of hemoglobinopathies

Hemoglobin	Mutation	Biochemical change	Classification	Clinical consequences
Hb Helsinki (rare)	Lys ^{82} β⇒Met	tetramer exhibits reduced binding of 2,3-bisphosphoglycerate	increased O_2 affinity	mild polycythemia (increased number of erythrocytes)
Hb Titusville (rare)	Asp ^{94} α ⇒Asn	heterodimer interface is altered to stabilize the T-state	decreased O_2 affinity	mild cyanosis (dark blue-purple skin coloration from deoxygenated blood)
HbM Boston (rare)	His ^{58} α ⇒Tyr	substitution of distal His causes structural perturbation in the heme pocket and decreased Bohr effect	ferric heme (methemoglobin)	cyanosis of skin and mucous membranes
HbS (common)	Glu ^{6} β ⇒Val	polymerization of deoxygenated protein occurs in microvasculature	altered solubility	hemolytic and vaso-occlusive crises; severe pain
Hb Gun Hill (rare)	Δ β$^{91–95}$	deletion of a critical Leu, required for heme contact, leads to misfolded protein	unstable protein	formation of Heinz bodies (inclusions of denatured hemoglobin); jaundice (yellow coloration of integument and sclera from heme metabolites); pigmented urine
Hb Agnana (rare)	β^{94} [+TG] ⇒ 156 amino acids	frameshift causes elongation to an unstable, nonsense protein	abnormal synthesis	severe anemia and other complications, consistent with a thalassemia (impaired synthesis of globin polypeptides)

HbC (Glu6β to Lys) cellular crystallation of oxygenated protein; increased fragility of rbcs
mild anemia and splenomegaly


Thank You!